

Technical Datasheet — VBF-T5R1NOCEILING-DSH-01

Cell: R4A MarginAnode1p0 (Chen2020 NMC811|graphite baseline + architecture overrides).

All values mechanically taken from the parameter set / simulation outputs; missing items honestly stated.

Field	Value	Source
Rated capacity (Ah)	9.6251 (1C discharge to 2.5 V)	cell/r4_A_calc_dfn.json:capacity_ah
Nominal voltage (V)	3.9016 (1C discharge midpoint)	cell/r4_A_calc_dfn.json:midpoint_voltage_v
Voltage window (V)	2.5 - 4.2	parameter set (Lower/Upper voltage cut-off)
Rated energy (Wh)	35.446	cell/r4_A_calc_dfn.json:energy_wh (time integration of $V \times I$ at 5 A)
Gravimetric energy density (Wh/kg)	583.93 (SPMe cross-check 583.46)	cell/r4_A_calc_dfn.json:energy_density_wh_kg (contract formula)
Volumetric energy density (Wh/L)	1042.10	cell/r4_A_calc_dfn.json:energy_density_wh_l
Maximum continuous discharge rate	1C (5 A nominal) simulated and clean to 2.5 V; rates above 1C not simulated	cell/r4_A_1c_dfn.json
Fast-charge capability	20 A (4C protocol; 2.08C of true capacity) accepted: 8.12 Ah in 1461 s to the 4.2 V ceiling at 45 degC ambient; no lithium plating (anode surface min +0.0212 V); T_max 320.36 K	cell/r4_A_4c_dfn.json
Operating temperature (simulated conditions)	1C discharge at 25 degC ambient; 4C charge at 45 degC ambient; T_max during 4C = 47.2 degC	protocol definitions; cell/r4_A_4c_dfn.json:T_max_K
Cycle life	Not simulated (requires aging model) — must not be treated as measured	honest omission
Safety determination	4C no-plating PASS (anode min +0.0212 V > 0); temperature PASS (320.36 K <= 333.15 K); abuse tests (nail / overcharge / crush) not simulated	cell/r4_A_4c_dfn.json; log evaluate R4A DFN
DC internal resistance (Ohm)	0.00109	cell/r4_A_calc_dfn.json:dcr_ohm
Power density (W/kg)	64930	cell/r4_A_calc_dfn.json:power_density_w_kg
Dimensions (mm)	65 (height) x 2370 (electrode width, single unwound layer) x 0.2208 (stack thickness); multilayer productization not modeled	parameter set + cell/r4_A_calc_dfn.json:thickness_m
Mass (g)	60.70 (electrolyte excluded per contract mass formula)	cell/r4_A_calc_dfn.json:mass_kg